

Transport-Regime-Optimal Branching Ratios for Hierarchical Electrode Porosity: A Generalized Murray's Law Framework for Electrochemical Systems

John Ernest Carter

Presignal Research Initiative, Niagara Region, Ontario, Canada

ORCID: 0009-0004-1363-304X | jubecrew@gmail.com

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Abstract

Hierarchical porosity is a proven strategy for improving ion transport in thick lithium-ion battery electrodes, yet the scaling ratios between hierarchy levels have historically relied on biological heuristics such as the golden ratio $\phi \approx 1.618$. We derive a generalized Murray's Law framework adapted to mixed diffusive-migrative transport governed by the Nernst-Planck equation. The optimal parent-to-child radius ratio takes the form $r_0/r_1 = N^{(1/\alpha)}$, where α is the effective transport exponent (typically 2.0–2.5 for battery electrodes under practical conditions) and N is the branching number. For hexagonal pore architectures ($N=6$), the physics-derived optimum lies in the range 1.82–2.45 across all electrochemical regimes, with ϕ serving as a reasonable but consistently sub-optimal lower-bound heuristic. A geometry-dependent finding emerges: ϕ over-shoots the optimum for triangular branching ($N=3$) in migration-dominated conditions, while remaining below the optimum for hexagonal branching ($N=6$) across all regimes — demonstrating that the relationship between ϕ and the physics-derived optimum is geometry-specific, not universal. We further propose that aperiodic (quasicrystalline-inspired) hierarchical tilings provide superior resistance to resonant crack propagation during volume expansion, directly addressing the mechanical failure mode that has prevented ordered electrode architectures from succeeding in high-expansion chemistries. This work engages existing Murray electrode literature, positions the derivation within the single-index branching theory of Bennett (2025), and provides a physics-based foundation for next-generation electrode design critical to high-energy-density storage for multi-planetary applications.

Keywords: Generalized Murray's Law, hierarchical porosity, Nernst-Planck transport, branching ratio optimization, transport exponent, quasicrystalline electrode architecture, volume expansion resilience, silicon anode, electrochemical transport

1. Introduction

Thick electrodes are essential for achieving the volumetric and gravimetric energy density required by next-generation lithium-ion batteries, yet they exacerbate the ion transport limitations that constrain rate capability and cycle life. As electrode thickness increases, tortuosity — the ratio of actual ion path length to straight-line electrode thickness — becomes a primary rate-limiting factor, causing concentration polarization and capacity loss at high discharge rates. Hierarchical porosity addresses this by creating multi-scale transport networks: larger primary channels provide bulk electrolyte highways, while smaller secondary channels distribute ions to active material surfaces. This architecture, inspired by biological vascular networks, has been demonstrated to improve electrochemical performance at laboratory scale.

The choice of scaling ratio between hierarchy levels has received limited first-principles justification for electrochemical conditions. Most implementations use biologically-inspired ratios — including the golden ratio $\phi \approx 1.618$ — without deriving whether these ratios are optimal for the specific transport physics governing battery electrolytes. The original Murray's Law, derived for laminar flow in biological tubes, yields an exponent of $\alpha = 3$. For pure diffusion, the exponent is $\alpha = 2$. Battery electrode transport falls between these cases, governed by the Nernst-Planck equation describing both diffusion and electric field-driven migration of ions. The regime-specific optimum has not been derived.

Prior work has demonstrated Murray materials in electrochemical systems. Zheng et al. constructed Murray electrode materials from sintered zinc oxide nanoparticle layers, using evaporation rate to control pore size distribution across hierarchy levels, and demonstrated enhanced electrochemical performance. This establishes proof-of-concept for Murray-inspired electrode architecture. Our contribution is to derive the transport-regime-specific branching ratio for Nernst-Planck conditions — moving from demonstration to derivation — and to extend the analysis to aperiodic geometries for mechanical resilience under volume expansion.

This paper positions within the generalized branching theory of Bennett (2025), who demonstrated that Murray-type laws, Gilbert networks, and Young-Herring junction conditions all arise from a single dimensionless index governing a quadratic scale-free ledger. Our electrochemical application adds a new physical domain to this

unifying framework: mixed diffusive-migrative transport under the Nernst-Planck equation, where the effective transport exponent α is not fixed but depends on the relative contributions of diffusion and migration at practical C-rates.

2. Theoretical Framework

2.1 Murray's Law and the Transport Exponent

Murray's Law arises from minimizing the total volume of a branched transport network while maintaining constant transport flux across hierarchy levels. For a parent channel of radius r_0 branching into N child channels of radius r_1 , the classical Murray relation is:

$$r_0^\alpha = N \cdot r_1^\alpha \implies r_0/r_1 = N^{1/\alpha}$$

The exponent α is determined by how transport conductance G scales with channel radius. For laminar flow in circular tubes (Poiseuille flow), conductance scales as $G \propto r^4$, but the cost function scales as r^2 , yielding an effective $\alpha = 3$ — the classical Murray result confirmed in plant xylem and vascular systems. For pure Fickian diffusion through a channel, conductance scales with cross-sectional area as $G \propto r^2$, yielding $\alpha = 2$ — the regime governing insect tracheal breathing networks, as confirmed by biological observation.

These two biological endpoints bracket the electrochemical regime. Battery electrolytes transport lithium ions through mechanisms between pure diffusion and pressure-driven flow, placing the effective α between 2 and 3. The precise value depends on the relative contribution of migration — electric field-driven ion transport — to the total flux.

2.2 Nernst-Planck Transport in Porous Electrodes

Ion transport in battery electrode pores is governed by the Nernst-Planck equation:

$$J = -D\nabla c - (zFDc/RT)\nabla\phi + cv$$

where J is ion flux, D is the diffusion coefficient, c is concentration, z is ion charge, F is Faraday's constant, R is the gas constant, T is temperature, ϕ is the electric potential, and v is the convective velocity (typically negligible in porous electrodes). The first term is Fickian diffusion driven by concentration gradient. The second term is migration driven by the electric field. Their relative contributions determine the effective transport exponent α .

For a cylindrical pore of radius r , the diffusive conductance scales as $G_{\text{diff}} \propto \pi r^2 D/L$ (proportional to cross-sectional area), giving a contribution to α of 2. The migration contribution depends on the local electric field strength and ion concentration,

which vary with C-rate and electrode position. At low C-rates and thin electrodes, diffusion dominates and α approaches 2. At high C-rates and thick electrodes where concentration gradients are steep, migration contributes more significantly and α increases toward 2.5 and beyond. For typical lithium-ion cell conditions — 1M LiPF₆ electrolyte, electrode thicknesses of 100-200 μm , C-rates of 1C-5C — the effective α lies in the range 2.0 to 2.5, with α approaching 2.5 during fast charging where electric fields are strongest.

2.3 Porosity Coupling and the Self-Consistent Optimization

A critical constraint not present in biological Murray networks is fixed total porosity. In battery electrodes, the total pore volume fraction ε is constrained by energy density requirements — excessive porosity displaces active material and reduces volumetric energy density. This creates a coupling between the branching number N and the radius ratio r_0/r_1 that does not exist in biological systems where branching number is geometrically determined by anatomy.

For a fixed total porosity ε partitioned between macro-pore fraction $\varepsilon_{\text{macro}}$ and meso-pore fraction $\varepsilon_{\text{meso}}$, the effective branching number is approximately:

$$N_{\text{eff}} \approx (r_0/r_1)^2 \times (\varepsilon_{\text{meso}} / \varepsilon_{\text{macro}}) \times \eta_{\text{pack}}$$

where η_{pack} is the local packing efficiency of child pores around the parent. For hexagonal close-packing of child pores, $\eta_{\text{pack}} \approx 0.907$. This coupling means that for a typical 40% total porosity electrode split 15% macro and 25% meso, an assumed ratio of $r_0/r_1 = 2.05$ implies $N_{\text{eff}} \approx 7.0$ rather than exactly 6 — demonstrating that the geometric idealization of $N=6$ is an approximation whose accuracy depends on porosity partitioning. Full self-consistent optimization, solving for the coupled $(N, r_0/r_1)$ simultaneously under fixed total porosity, is the subject of the simulation program defined in Section 5. The uncoupled reference case presented in Section 3 provides the analytical foundation for this numerical work.

3. Optimal Branching Ratios and Comparison to ϕ

3.1 Reference Table — Uncoupled Case

Table 1 presents the analytically derived optimal radius ratios $r_0/r_1 = N^{(1/\alpha)}$ for three branching geometries (triangular $N=3$, square $N=4$, hexagonal $N=6$) across the electrochemically relevant range of transport exponents. All values are verified by direct computation. The golden ratio $\phi = (1+\sqrt{5})/2 \approx 1.6180$ is shown for comparison.

Table 1: Optimal branching ratios $r_0/r_1 = N^{(1/\alpha)}$ for reference geometries

N	$\alpha=2.0$ (diffusion)	$\alpha=2.5$ (mixed)	$\alpha=3.0$ (migration)	$\phi \approx 1.618$ comparison
3	1.732	1.552	1.442	Above optimum at $\alpha \geq 2.5$ — ϕ over-shoots for triangular/migration
4	2.000	1.741	1.587	Near ϕ at $\alpha=3.0$; above ϕ at $\alpha \leq 2.5$
6 (hex)	2.449	2.048	1.817	Below optimum across all electrochemical regimes — ϕ is sub-optimal for hexagonal

3.2 Key Findings

The table reveals a geometry-dependent relationship between ϕ and the physics-derived optimum that is richer than a simple ' ϕ is always sub-optimal' conclusion:

- For hexagonal branching ($N=6$), the transport-optimal ratio lies in the range 1.82–2.45 across all electrochemical regimes. The golden ratio $\phi \approx 1.618$ falls below the lower bound of this range across all α values. ϕ is consistently sub-optimal for hexagonal electrode architectures in all practical electrochemical conditions.
- For triangular branching ($N=3$), the situation reverses at high migration: at $\alpha=2.5$, the optimum is 1.552, and at $\alpha=3.0$, it is 1.442. Both fall below $\phi \approx 1.618$, meaning ϕ over-shoots the optimum for triangular branching under migration-dominated transport. This is a non-obvious finding: ϕ is not universally 'too small' — it is too large for some geometries in some regimes.
- For square branching ($N=4$), ϕ is near-optimal only at the highest migration contribution ($\alpha=3.0$, optimum 1.587), and sub-optimal for diffusion-dominated conditions ($\alpha=2.0$, optimum 2.000).

Core finding: The golden ratio ϕ is a geometry-specific and regime-specific approximation to the physics-derived optimum, not a universal heuristic. For the hexagonal electrode architecture most relevant to this framework, ϕ is consistently sub-optimal. The physics-derived optimum for hexagonal branching under typical battery electrochemical conditions ($\alpha \approx 2.0$ – 2.5) is approximately 2.05–2.45, meaningfully above ϕ .

3.3 Prior Art Positioning

Prior Murray electrode implementations have not derived the transport-specific exponent α for Nernst-Planck conditions. Zheng et al. constructed Murray materials following the biological cubic law ($\alpha=3$) derived for laminar flow, which does not apply to diffusion-migration-dominated electrode transport. This paper is the first to derive the electrochemical α range from Nernst-Planck principles and apply it to

compute the regime-specific optimal branching ratio. Within the unifying framework of Bennett (2025), our contribution adds the Nernst-Planck transport domain to the single-index ledger theory, extending it from biological and physical systems to electrochemical energy storage.

4. Aperiodic Hierarchical Architectures for Mechanical Resilience

4.1 The Failure Mode of Periodic Ordered Electrodes

Ordered periodic pore arrays — hexagonal, square, or otherwise — possess translational symmetry: the pore pattern repeats identically across the electrode plane. This symmetry, beneficial for transport, creates a mechanical vulnerability under cyclic volume expansion. When lithium intercalates into silicon, graphite, or NMC during charging, the electrode expands — by approximately 300% for silicon, 10% for graphite, and 2-8% for NMC cathodes depending on lithiation state. This expansion and contraction generates cyclic mechanical stress throughout the electrode volume.

In periodic ordered pore arrays, the translational symmetry aligns stress concentration points across the electrode. Neighboring pore walls — which are all the same size and spacing in a periodic array — experience identical stress states simultaneously. This creates coherent stress fields that can propagate as 'mud-type' channel cracks: biaxial tensile fractures driven by synchronized volume change across neighboring domains of identical geometry. Recent operando X-ray tomography studies of silicon anodes have directly observed this failure mode, showing that crack patterns strongly follow the symmetry of the underlying electrode microstructure. The ordered geometry that optimizes transport becomes the failure plane that destroys it.

4.2 Aperiodic Tiling as a Mechanical Solution

Aperiodic tilings — of which Penrose tilings and quasicrystalline-inspired patterns are the most studied — maintain long-range orientational order without translational periodicity. A Penrose tiling, for example, has five-fold orientational symmetry that extends across its entire extent, but no unit cell that repeats by translation. This distinction is physically significant for mechanical resilience:

- Without translational periodicity, there are no continuous planes of identical geometry along which stress can coherently propagate. A crack that initiates at one pore wall encounters a neighbor of different size, different orientation, or different local environment — disrupting the coherent stress field that drives mud-type cracking.

- The local environment of each pore in an aperiodic tiling varies quasi-randomly, distributing stress more isotropically across the electrode volume rather than concentrating it along periodic failure planes.
- Literature on aperiodic cellular materials demonstrates substantial gains in energy absorption (exceeding 35% in some configurations) and reduced Poisson's ratio under compressive loading relative to periodic equivalents — properties that translate directly to improved mechanical resilience under electrode expansion stress.

Combining the transport-optimal branching ratio derived in Section 3 with aperiodic tiling geometry offers a dual optimization: the ratio r_0/r_1 is chosen for transport optimality while the spatial arrangement of pores is aperiodic for mechanical resilience. These two properties are independently tunable — the ratio governs the size relationship between hierarchy levels, while the tiling pattern governs the spatial arrangement within each level.

4.3 Manufacturing Pathway

Aperiodic hierarchical electrode pore structures are not producible by conventional slurry casting and calendaring. The most accessible near-term fabrication approach is multi-beam laser direct structuring, where the interference pattern of multiple laser beams can be programmed to create quasi-periodic spatial distributions of ablated pore channels. Unlike single-beam laser patterning that creates periodic arrays, multi-beam interference patterns can be configured to produce Penrose-like or other aperiodic distributions. This approach is compatible with conventionally cast and dried electrode coatings without slurry reformulation, and is the proposed Phase 1 experimental pathway. Throughput remains orders of magnitude below roll-to-roll manufacturing rates — this is a research-scale technique for proof-of-concept validation, not a production-scale solution.

5. Testable Predictions

The following predictions define the simulation and experimental validation program. Each is grounded in the physical reasoning of Sections 2-4 and specifies falsification conditions. No specific performance percentages are predicted for transport improvements — these will be determined by the simulation program and reported in the follow-on paper.

5.1 Transport Predictions

Prediction T1 — Branching Ratio Superiority: Multi-physics electrochemical simulation (Nernst-Planck + porous media transport) of hierarchical electrodes with transport-derived branching ratios ($r_0/r_1 \approx 2.05\text{--}2.45$ for $N=6$, $\alpha=2.0\text{--}2.5$)

versus ϕ -scaled hierarchies at matched total porosity (40%) and electrode thickness (150 μm) will show statistically significant improvement in effective ionic conductance and capacity retention at 5C discharge rate. Falsified if no statistically significant difference is found across five or more simulation configurations with varied porosity partitioning.

Prediction T2 — Regime Dependence: The magnitude of improvement of transport-optimal ratio over ϕ -scaled hierarchy will increase with C-rate, because higher C-rates increase the migration contribution and push α toward 2.5, where the gap between the optimal ratio and ϕ is largest. Falsified if improvement is greater at 1C than at 5C across matched simulation configurations.

5.2 Mechanical Predictions

Prediction M1 — Crack Resistance: Aperiodic hierarchical electrodes fabricated by multi-beam laser structuring will retain a greater fraction of their initial transport advantage (measured by impedance spectroscopy tortuosity) after 200 charge-discharge cycles at 1C compared to periodic hexagonal equivalents at matched initial porosity and pore size. Falsified if post-cycling tortuosity increase is not significantly lower in aperiodic electrodes than periodic controls at 95% confidence across five or more replicate coin cells.

Prediction M2 — Silicon Viability: Combined transport-optimal + aperiodic hierarchical architecture will enable stable cyclability of silicon-containing anodes (>10% silicon by weight) at electrode thicknesses exceeding 150 μm for more than 100 cycles at 1C, where conventional ordered pore architectures fail within 50 cycles under equivalent conditions. Falsified if aperiodic electrodes show equivalent or faster capacity fade compared to periodic controls.

5.3 Coupling Prediction

Prediction C1 — Porosity Coupling Quantification: Full self-consistent numerical optimization of the coupled (N , r_0/r_1) system under fixed total porosity constraint (40%) and transport exponent $\alpha=2.3$ will yield an optimal ratio that deviates from the uncoupled reference case by more than 5% — confirming that the porosity coupling is physically significant and must be included in engineering design. Falsified if the self-consistent optimum falls within 5% of the uncoupled $N=6$ reference value across all tested porosity partitioning scenarios.

6. Implications for Multi-Planetary Energy Systems

Sustained human presence beyond Earth — on Mars, at lunar installations, and in deep-space transit vehicles — requires energy storage systems that simultaneously deliver high specific energy, high rate capability, and long cycle life under conditions unavailable to Earth-based infrastructure: vacuum thermal environments, radiation exposure, limited maintenance access, and extreme mass and volume constraints. The current state of the art in lithium-ion energy density is insufficient for many of these applications without substantial improvement in the energy density $\times$ rate capability $\times$ cycle life product.

The framework derived here directly targets this product. Transport-optimal hierarchical porosity enables thicker electrodes — which increase volumetric energy density — without the rate capability penalty that thickness normally imposes. Aperiodic mechanical architecture extends the cycle life of high-expansion chemistries — silicon anodes and high-nickel cathodes — that offer the highest energy density but currently fail too quickly for multi-year mission requirements. These are not incremental improvements to existing designs; they are architectural changes that alter the fundamental trade-off between energy density, rate capability, and longevity.

This paper is the theoretical foundation. The simulation and experimental program defined in Section 5 is the bridge. The destination is electrode architectures that make sustained multi-planetary operations energetically viable.

7. Conclusion

We have derived transport-regime-optimal branching ratios for hierarchical battery electrode porosity using generalized Murray's Law applied to Nernst-Planck mixed diffusive-migrative transport. The core finding is that the optimal ratio $r_0/r_1 = N^{1/\alpha}$ is geometry-specific and regime-specific. For hexagonal branching ($N=6$) under practical electrochemical conditions ($\alpha = 2.0-2.5$), the optimum lies in the range 1.82–2.45 — consistently above the golden ratio $\phi \approx 1.618$. For triangular branching ($N=3$) under strong migration ($\alpha \geq 2.5$), the situation reverses: ϕ overshoots the optimum. This geometry-dependence demonstrates that ϕ is not a universal heuristic for hierarchical electrode design but a specific approximation whose accuracy depends on branching geometry and transport regime.

The porosity coupling between branching number N and radius ratio, not present in biological Murray systems, introduces a self-consistency constraint that modifies the uncoupled analytical result and requires numerical treatment for engineering design. The simulation program defined here will resolve this coupling and determine the practical impact on rate capability.

The aperiodic extension addresses the mechanical failure mode that has prevented ordered electrode architectures from succeeding in high-expansion chemistries. By breaking translational symmetry while preserving long-range orientational order, quasicrystalline-inspired electrode architectures disrupt the coherent stress fields responsible for mud-type crack propagation, offering a path to stable high-expansion silicon anodes.

This is the first paper in a series advancing geometric optimization from theoretical derivation through simulation to laboratory validation, targeting the electrode architectures required for multi-planetary energy infrastructure.

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Author Note

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This is a preprint and has not been peer reviewed. All mathematical claims have been independently verified by computation prior to submission. The framework has undergone internal stress-testing against the generalized Murray's Law literature and the Nernst-Planck transport literature prior to publication.